

HOSTEL SECURITY MONITORING SYSTEM



FACE RECOGNITION

REAL-TIME ALERTS

LIVE MONITORING

SECURE DATA STORAGE



Dashboard

Live View

Students

Entry / Exit Logs

Alerts

Reports

Settings

Total Students

128

Inside Hostel

85

Outside Hostel

43

Live Camera

Recent Entry / Exit

ID	Name	Type	Time
21CS042	SUGAPRIYA V	Entry	09:15 AM
21CS017	THANAVATHI P	Exit	09:10 AM
21CS038	HEMASRI S	Entry	09:05 AM
21CS021	KAVIYA R	Exit	08:55 AM
21CS011	NITHYA S	Entry	08:45 AM



Team Name:
TEAM TRACKERS

Team members



SUGAPRIYA V



THANAVATHI P



HEMASRI S



PROBLEM STATEMENT

AI BASED HOSTEL SECURITY MONITORING SYSTEM

In most hostels, the entry and exit of students are managed manually. This traditional method has several limitations and security risks.



MANUAL RECORD KEEPING

Entry and exit are recorded in registers manually which is time-consuming and prone to human errors.



NO REAL-TIME MONITORING

There is no system for real-time monitoring of students going out or returning to the hostel.



NO INSTANT ALERTS

Wardens and parents are not notified immediately when a student leaves the hostel.



DATA MANAGEMENT ISSUES

Manual data is difficult to manage, search and maintain for long-term reference.



SECURITY CONCERNS

Unauthorized access and lack of proper verification can lead to security risks for students.



Manual Entry in Registers



No Real-time Monitoring



No Instant Alerts



Hard to Manage Data



Security Risks



There is a need for an automated system that can monitor, verify and alert in real-time to ensure better security and safety.

OBJECTIVES

HOSTEL SECURITY MONITORING SYSTEM

The main objectives of our project are:



01

ENHANCE SECURITY

To enhance the security of hostel students by monitoring entry and exit in real-time.



02

AUTOMATIC IDENTIFICATION

To use AI-based face detection and recognition to automatically identify students.



03

INSTANT ALERTS

To send instant alerts and notifications to wardens and parents when a student leaves the hostel.



04

DATA MANAGEMENT

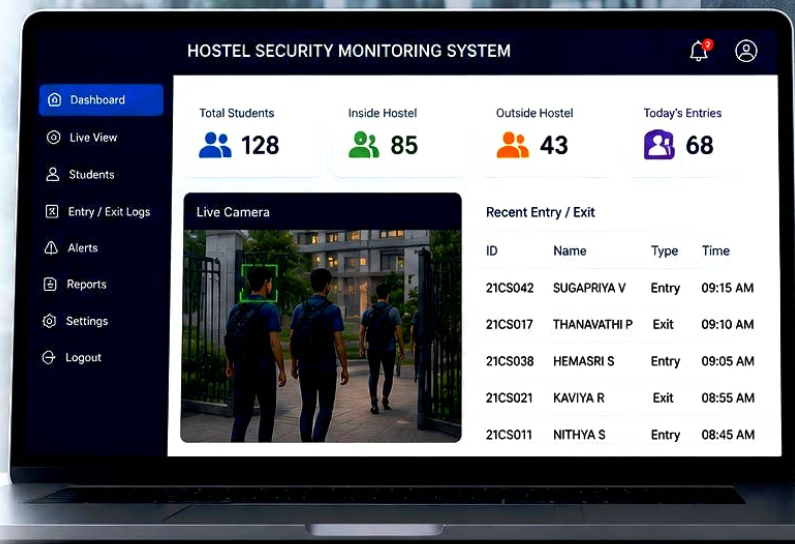
To store and manage all entry/exit records securely for future reference and tracking.



05

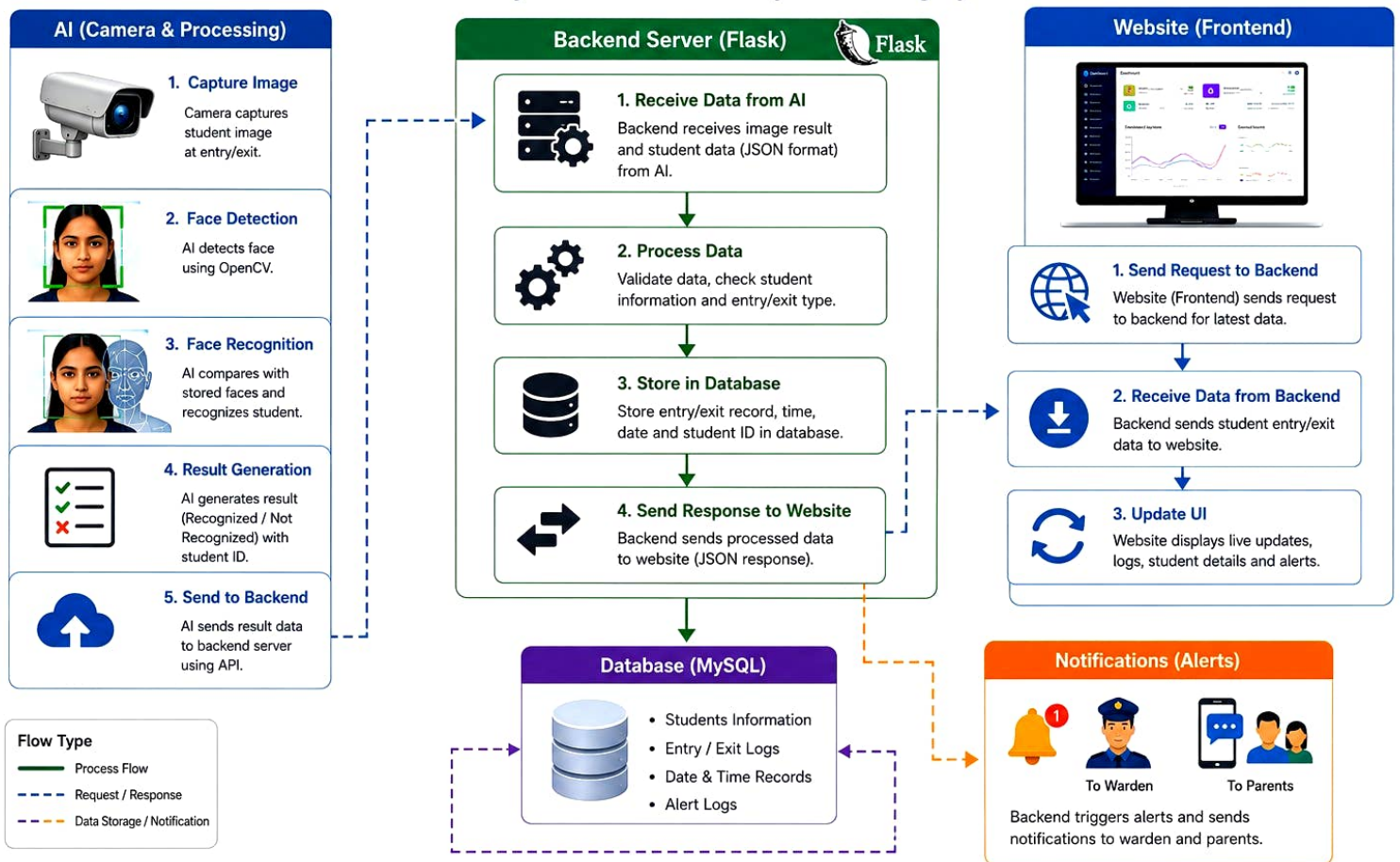
REAL-TIME MONITORING

To provide a web dashboard for real-time monitoring and management of student movements.



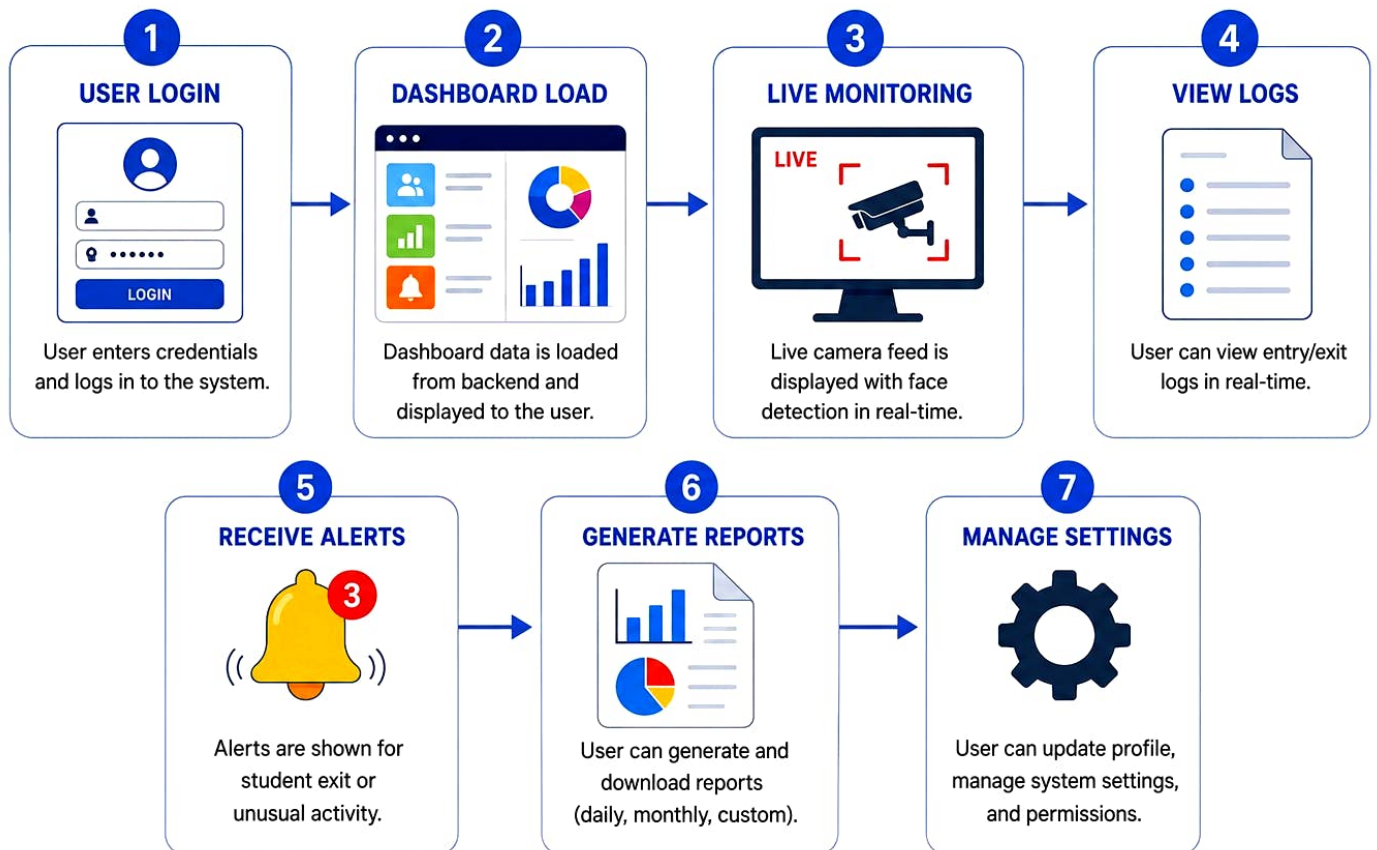
Communication Between AI and Website

In Our Project - Hostel Security Monitoring System



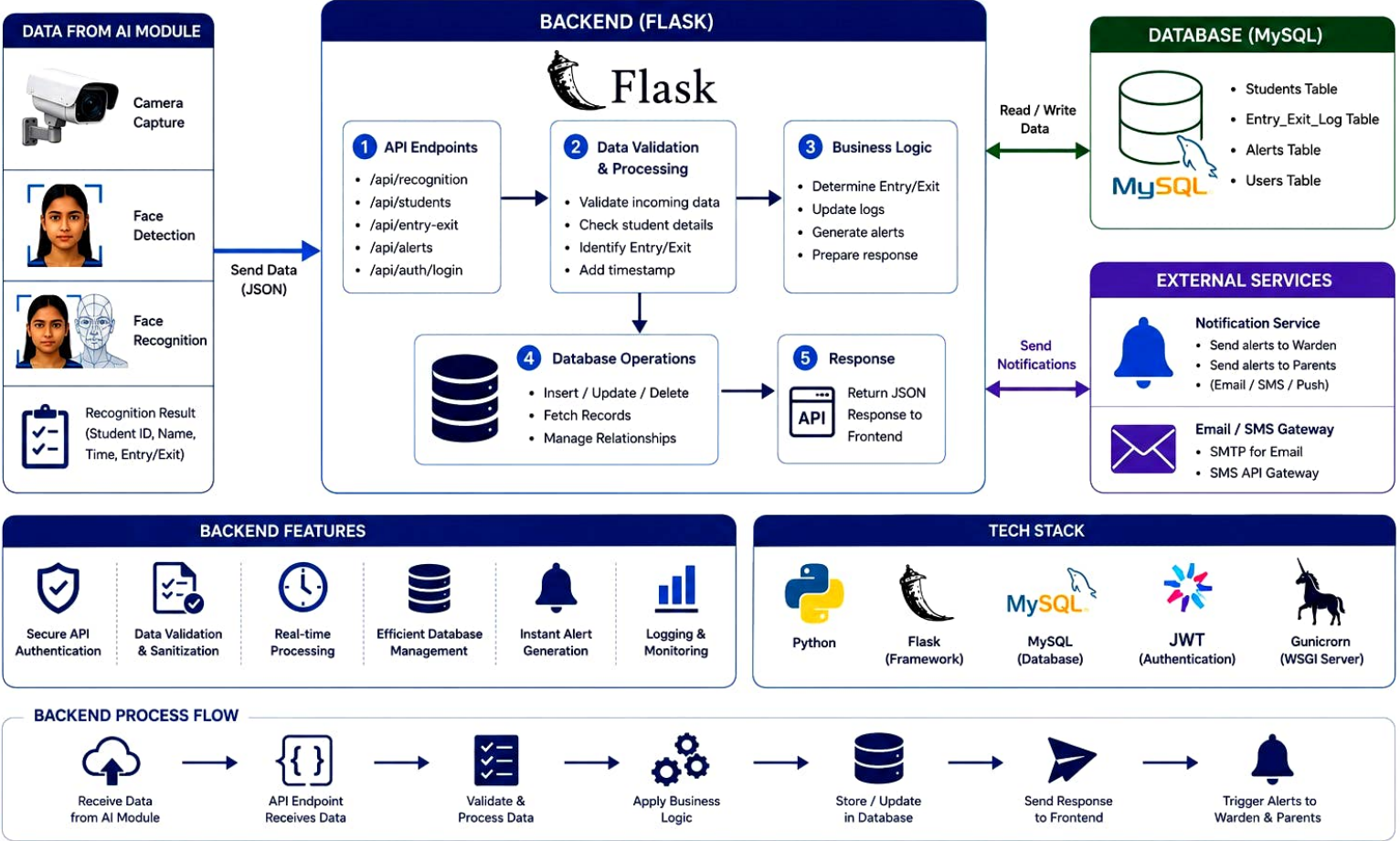
FRONTEND PROCESS

HOSTEL SECURITY MONITORING SYSTEM



BACKEND DEVELOPMENT

Hostel Security Monitoring System



BACKEND PROCESS

HOSTEL SECURITY MONITORING SYSTEM

1. USER INTERFACE

Inside Hostel
85

Security Alerts
07

Recent Activity

ID	Name	Type	Time
21CS042	Sugapriya V	Entry	09:15 AM
21CS017	Thanavathi P	Exit	09:10 AM
21CS038	Hemasri S	Entry	09:05 AM
21CS021	Kaviya R	Exit	08:55 AM
21CS011	Nithya S	Entry	08:45 AM

[View All](#)

Live Logs

ID	Name	Type	Gate	Image
21CS042	Sugapriya V	Entry	Gate-1	
21CS017	Thanavathi P	Exit	Gate-1	
21CS038	Hemasri S	Entry	Gate-1	
21CS021	Kaviya R	Exit	Gate-1	
21CS011	Nithya S	Entry	Gate-1	

Notifications

Notifier sent to Parents
Student 21CS042 has left the hostel.
Notified to Warden & Parents.

2. BACKEND (SYSTEM LOGIC & DATABASE)

1. System Architecture

2. Backend Flow

3. Database Structure (MySQL)

students

- id (PK)
- name
- roll_no
- course
- year
- phone
- email
- photo

entry_exit_log

- id (PK)
- student_id (FK)
- date
- time
- type (entry/exit)
- gate
- image

alerts

- id (PK)
- student_id (FK)
- message
- alert_type
- status
- timestamp

3. UPS (POWER BACKUP)

1. What is UPS?

UPS (Uninterruptible Power Supply) provides backup power to the system when main power fails. It ensures continuous operation of server and devices.

2. UPS Connection Diagram

3. Benefits

- ✓ Provides backup during power failure
- ✓ Protects system from sudden shutdown
- ✓ Ensures data safety
- ✓ Improves system reliability

DATABASE DESIGN

AI Hostel Security Monitoring System

DATABASE TABLES

1. students

Field Name	Type	Description
id (PK)	INT	Unique Student ID
name	VARCHAR	Student Name
roll_no	VARCHAR	Roll Number
department	VARCHAR	Department
phone	VARCHAR	Phone Number
email	VARCHAR	Email ID
face_encoding	TEXT	Encoded Face Data
created_at	DATETIME	Record Created Time

2. entry_exit_log

Field Name	Type	Description
id (PK)	INT	Unique Log ID
student_id (FK)	INT	Reference to Student
entry_exit_type	ENUM	Entry / Exit
gate_id	VARCHAR	Gate Number
timestamp	DATETIME	Date & Time
image_path	VARCHAR	Captured Image Path

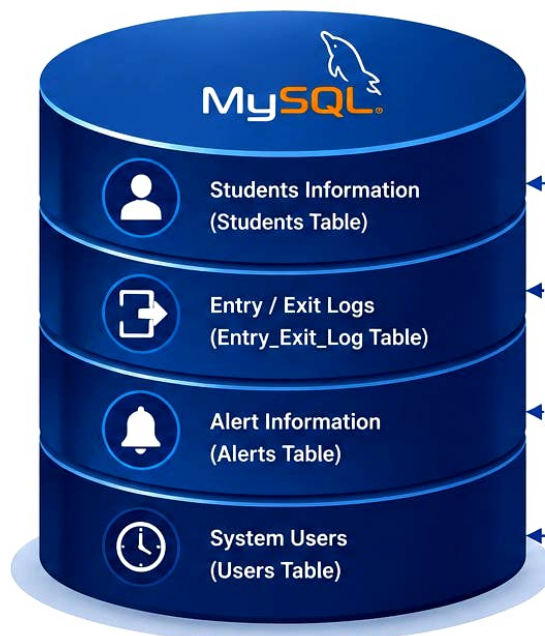
3. alerts

Field Name	Type	Description
id (PK)	INT	Unique Alert ID
student_id (FK)	INT	Reference to Student
message	TEXT	Alert Message
sent_to	ENUM	Warden / Parent
timestamp	DATETIME	Alert Time
status	ENUM	Sent / Pending

4. users

Field Name	Type	Description
id (PK)	INT	Unique User ID
username	VARCHAR	Username
password	VARCHAR	Password
role	ENUM	Admin / Warden / Security
created_at	DATETIME	Record Created Time

DATABASE (MySQL)



RELATIONSHIP KEY



Primary Key (PK)



Foreign Key (FK)

- - - - - Relationship

SYSTEM FLOW



1. Camera Captures Image



2. Face Detection & Recognition



3. Data Sent to Backend Server



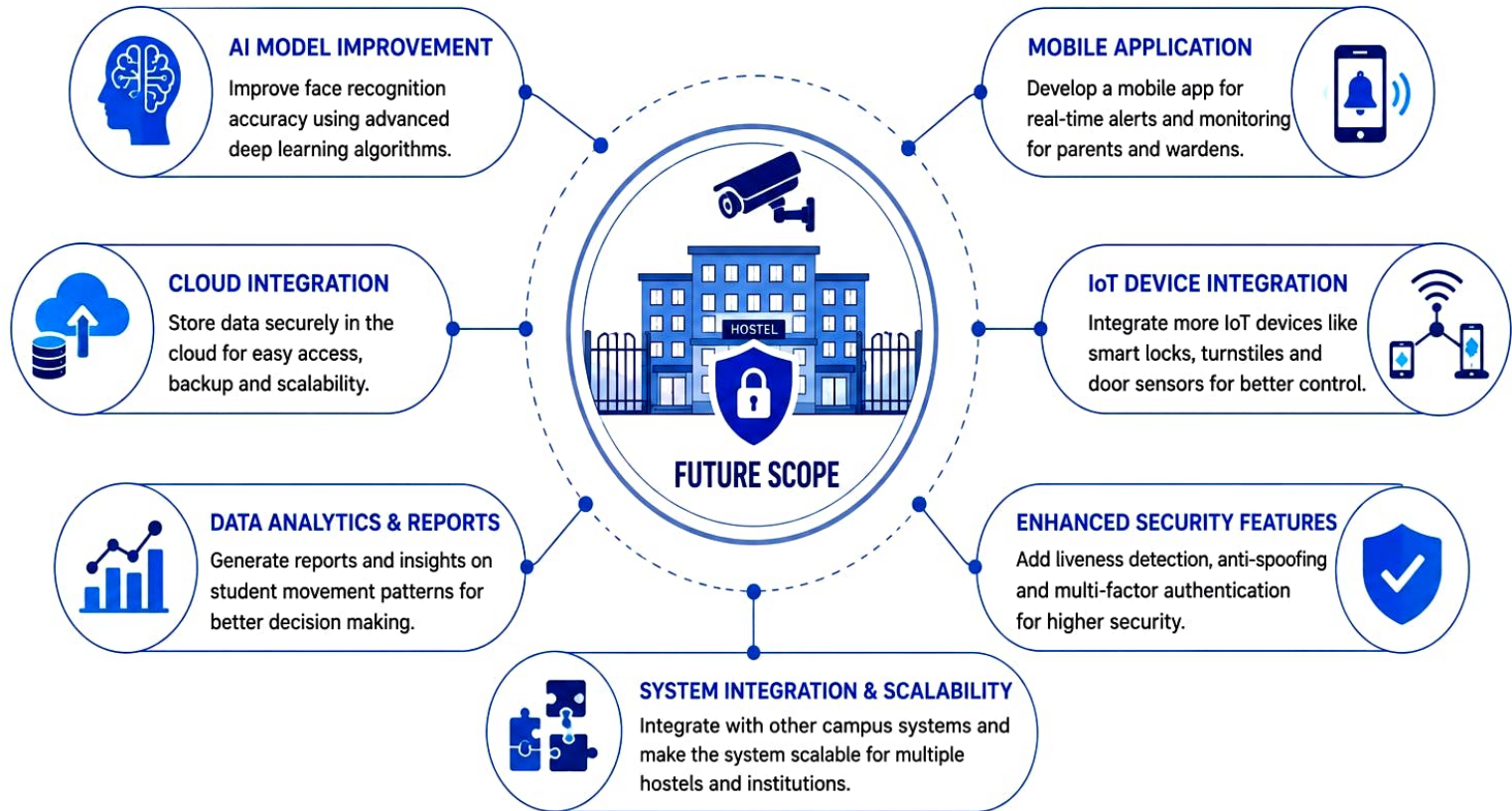
4. Data Stored in Database



5. Data Fetched & Shown in Dashboard

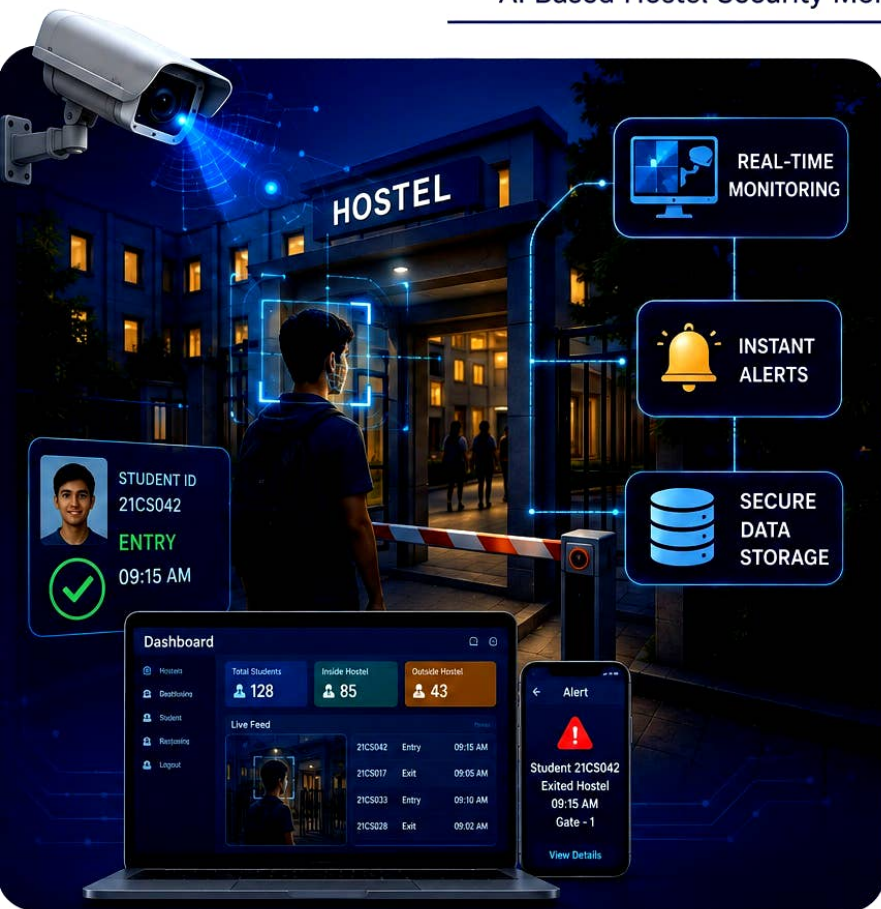
FUTURE SCOPE

HOSTEL SECURITY MONITORING SYSTEM



ADVANTAGES

AI Based Hostel Security Monitoring System



IMPROVED SECURITY

Enhances safety of students by automating entry and exit monitoring.



REAL-TIME MONITORING

Continuous live monitoring of hostel entrances and exits.



REDUCES MANUAL WORK

Automates the process and reduces manual register maintenance.



ACCURATE DIGITAL RECORDS

Stores all entry and exit data accurately in the database.



INSTANT ALERTS

Sends real-time notifications to wardens and parents.



SECURE DATA STORAGE

All data is securely stored and protected from loss.



SAVES TIME & IMPROVES EFFICIENCY

Speeds up the process and improves overall efficiency.